

BMDS2114 Machine Learning Assignment

Aims	1. Enable students to analyze and employ appropriate Machine Learning techniques to solve problems. 2. Enable students to use relevant tools and technology, such as python programming, to visualize and evaluate the performance of different machine learning algorithms.
Learning Outcomes Assessed	1. Justify the machine learning methods and strategies for the data set based on the performance evaluation. (CLO2, A3, PLO5) 2. Produce machine learning programs using programming language or other available technologies to solve a real world problem. (CLO3, P4, PLO3)
Outline of Problem	For your assignment, you should critically evaluate current technologies, propose a project for the selected machine learning topic, and implement a machine learning solution to solve the problem in the proposed project by using Python.
Details	Your team should consist of three to four (3-4) members. Your team must propose an appropriate assignment title. Each group must propose a unique title or solution; duplication of ideas is not permitted. Please be aware that the assignment's basic requirements are outlined below. Meeting these requirements may result in an Average or Good grade. To achieve an Excellent grade, additional efforts are necessary, such as acquiring new skills, introducing innovative ideas, implementing complex machine learning algorithms, showcasing proficiency in handling big data, and/or delivering outstanding reports accompanied by a functional prototype.
Machine Learning Project Checklist	This checklist can guide you through your Machine Learning projects. The main steps include: 1. Frame the problem and define the approach. 2. Get or define the data and/or environment. 3. Explore the data and/or environment dynamics. 4. Prepare the data or state representations. 5. Select and implement appropriate models or algorithms. 6. Train and evaluate models or agents. 7. Fine-tune parameters and optimize performance. 8. Interpret, visualize and present results.

Project Details Part A – Identifying and Selecting Promising Models	1. Train a variety of diverse models or learning approaches using different methodologies suited to the problem. - Apply different algorithms to capture underlying patterns and relationships within the data. - Consider models that are well-suited for structured data, pattern recognition, sequential decision-making, or adaptive learning based on interaction. 2. Measure and compare performance using relevant evaluation techniques. - Apply appropriate validation methods, such as cross-validation, bootstrapping, or adaptive testing. - Use metrics suited to the task, such as accuracy, precision-recall, stability of learned strategies, or pattern separation measures. - Analyze robustness and reliability across different data distributions and conditions. 3. Analyze significant patterns and influencing factors. - Identify key variables, underlying structures, or decision boundaries that impact learning performance. - Use interpretability tools, visualizations, and statistical methods to understand how information is processed and utilized. 4. Examine types of errors or inefficiencies. - Investigate incorrect predictions, inconsistencies in identified patterns, or inefficiencies in decision-making. - Evaluate generalization ability, adaptability, and potential biases affecting learning outcomes. - Assess areas where improvements can be made through better representations, enhanced exploration, or refined adjustments. 5. Refine feature selection, data representations, and structural configurations. - Optimize information processing by transforming inputs, improving representations, or restructuring learning procedures. - Adjust learning mechanisms to ensure meaningful differentiation, adaptability, and efficiency. 6. Iterate through the previous steps, incorporating insights gained from exploratory analysis, refinements, tuning strategies, and validation results.
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	7. Shortlist the most promising models or strategies. - Select approaches based on effectiveness, interpretability, and adaptability to different scenarios. - Choose models that provide meaningful insights, efficient learning, and reliable outcomes. Note: Aim to automate these steps as much as possible to enhance efficiency and scalability across different problem domains.
Project Details Part B – Refining the model/system	1. Improve how the model learns by adjusting its settings - Fine-tune different options that control how the model processes data and makes decisions. - Try different ways of handling missing or unclear information, such as filling in gaps with estimated values, using common patterns in the data, or removing incomplete entries. - When testing different settings, randomly exploring options can be more efficient than checking every possibility. For models that take longer to train, smarter search methods can help find the best settings faster. 2. Combine different models to improve accuracy and reliability - Using multiple models together can lead to better overall results. - Consider blending different models to take advantage of their strengths, which can help reduce mistakes and make the model/system more effective. 3. Test the model in different situations to check its reliability - Measure how well the model performs on new data to make sure it works consistently. - Look at the types of mistakes it makes and think about how to reduce them. - If the system is meant to handle changing information, test whether it adapts well over time. 4. Make the process more efficient and scalable - Automate repetitive tasks like fine-tuning settings, picking the most important pieces of data, and evaluating performance. - In cases where data is more complex, using advanced models may help recognize patterns more effectively. Note: try to automate these steps as much as possible. You may also make use of deep learning model to solve the problem. You are required to include proper comments and remarks in your Jupyter Notebook along with your coding to explain and justify the system outcomes.

AI Use Category for This Assignment	This assignment is classified as: ● Yellow — Limited AI: Students may use AI tools for brainstorming, concept clarification, syntax support, data analysis, language refinement, and technical support. However, they must not be used to generate core arguments, critical analysis, or original academic content. The use of AI to generate complete solutions, produce reports, or conceal lack of contribution is strictly prohibited. All use of AI tools must be declared by completing the prescribed AI Usage Disclosure Form. Any AI-assisted work must be verified and adapted to reflect the team's actual assignment. Each student must be able to explain, justify, modify, and demonstrate all submitted work during presentations. Undeclared or inappropriate use of AI may result in penalties, including a reduction in marks.
Submission Deadlines	Final Report: Week 11
Academic Integrity and Plagiarism	There must be originality in your work, i.e. do not copy directly from other group(s). You may work with your team member(s) and discuss with your tutor to produce the solution of this assignment. Before submitting your assignment, please make sure that you have complied with TARMT plagiarism Policy. Any cheating, attempt to cheat, plagiarism, collusion and any other attempts to gain an unfair advantage in assessment will cause the students concerned to be penalized. IMPORTANT: Students found to be dishonest are liable to disciplinary action.
Late Submission	Late submission without valid reason will NOT be tolerated. For late submission, there will be a reduction of total marks: • Late 1 to 3 days after deadline of submission: minus 10 marks • Late 4 to 7 days after deadline of submission: minus 20 marks • Late more than 7 days after deadline of submission: 0 marks In certain circumstances, a student may be allowed to submit the assignment late with valid reason. S/he must contact the tutor at least one week before the assignment is due. The tutor will evaluate whether the circumstance warrants submitting the assignment late, but no guarantee that the students will not be penalized. Failed to submit the reports & code will lead to failure of the coursework.

Project Details: Final Report Writing

What to hand in?	Students are required to come out with the final report for the selected machine learning topic. Project assessment (demo, question and answer) will be carried out during the practical class starting week 12. The final report should contain the following: • Title of the proposed project. • Brief description of your proposed machine learning solution to solve the problem. (Problem Statement) • Describes how your solution works on a level that does not lose important details. Make some sensible breakdown into headings (Methodology). • Includes and explain the evaluation outcome in the article (Result). • Use clear, well written English language. • Task allocations for each of the respective group member and a project plan. ** Each student is expected to produce individual work regarding the specified items. • The report shall be compiled and submitted in group.
Format for Deliverable	Refer to the guidelines below for the assignment report, any additional guidelines will be provided by the tutor. 1. Paper size: A4 2. Font/spacing: Times New Roman, 11 points, single spacing 3. Cover page 4. Assessment rubric 5. Around 40 pages (below 50 pages) Note: Submit all your materials (final report and source code) to the Google Classroom assignment submission track.